

A Covariant Action for the Elastic Dark-Energy Medium

The Branch-B field theory written: a relativistic two-invariant elastic solid whose volumetric sector carries the $a_0 = c^2 \sqrt{\Lambda/32\pi}$ displacement law and terminates at the derived entropy-budget cutoff $y_c = Z/2$ — turning the posited shear fraction w and the Cassini quadrupole into defined computations

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Abstract

The elastic-medium branch (Branch B) of the de Sitter–Unruh a_0 program has until now been a constitutive prescription without an action — its structural debt (P1). This note writes the action. The medium is a relativistic elastic solid in the standard three-scalar formulation [Carter & Quintana 1972; Bucher & Spergel 1999; Battye & Moss 2007; Endlich, Nicolis & Wang 2013]: fields $\varphi^a(x)$, $a = 1, 2, 3$, internal metric $B^{ab} = g^{\mu\nu} \partial_\mu \varphi^a \partial_\nu \varphi^b$, with the medium’s rest frame, number density, volumetric strain J , and deviatoric (shear) strain Σ all built from B^{ab} . The Lagrangian is

$$S = \int \sqrt{-g} \left[\frac{c^4 [1 + h(J)] R}{16\pi G} - \rho_\Lambda c^2 \mathcal{G}(\Sigma) \right] + S_{\text{baryons}}[g, \psi], \quad \mathcal{E} = 1 + \frac{V(J) + \mu_s \Sigma}{\rho_\Lambda c^2},$$

where the relaxed medium is the dark energy ($\mathcal{E}(0, 0) = 1$), and the constitutive content — all of it inherited from previously machine-verified results, none newly fitted — is: **(i)** a volumetric potential $V(J)$ whose deep-regime branch is the scale-free law $V'(J) \propto \sqrt{J}$ with total deep stiffness $K_{\text{eff}} = a_0^2/16\pi G$, fixed by matching the Verlinde displacement response $g_D = \sqrt{a_{0V} g_{\text{bar}}/6}$ at $a_{0V} = cH_\Lambda = Za_0$ (the match that lands at $\sqrt{Z/6} = 0.982$ of the required lensing source); **(ii)** a domain boundary at $|J| = J_c$ implementing the entropy-budget constraint $S_M \leq S_{DE}$, whose location is the derived cutoff $y_c = Z/2$ (pinned to Verlinde’s published coefficients) — the budget lives outside the constitutive law, exactly as the single-valued- F impossibility theorem requires; **(iii)** a linear shear sector with modulus μ_s , bounded above by the exact refusal $\rho_\Lambda c^2/K_{\text{eff}} = 6Z^2 = 201.1$ (the medium’s shear waves are material sound at $v_T = \sqrt{\mu_s/\rho_\Lambda} \leq 0.17c$, not gravitational waves — metric tensor modes propagate at c untouched, so GW170817 is satisfied in every sector by construction, and light bends by the medium’s genuine stress-energy in the one shared metric). **Corrected in v2 (a self-caught overclaim):** the *minimally coupled* solid cannot gravitationally source the halo — its strain energy falls short of the required apparent-dark-matter density by $10^{6.5-7}$ (computed; the v1 script never actually checked this item). The mechanism is therefore **nonminimal**, as in Verlinde’s own construction: the volumetric strain modulates the gravitational response through $[1 + h(J)]R$, with h fixed at the weak-field level by $\nabla^2 h = (8\pi G/c^2)\rho_D$ — giving $h \sim 10^{-6}$ everywhere (weak), deep-Newton-suppressed at high accelerations ($\nabla h \rightarrow 2(\nu - 1)g/c^2 \rightarrow a_0/c^2$), and coupled to the *trace* sector only, so the two-invariant monopole/quadrupole decoupling and the shear-fraction analysis are preserved unchanged. Machine-verified consequences of the corrected action: the nonminimal weak-field limit reproduces the required apparent-dark-matter source (now checked, not asserted); the background cosmology

returns $\rho_{\text{DE}} = \rho_\Lambda$ at the relaxed point with $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\text{DE}}(z)}$ inherited automatically (the falsifiable $a_0(z) \propto \sqrt{\rho_{\text{DE}}}$ evolution is now a property of the action, not a separate posit); the phonon spectrum is ghost-free and stable for $V'' > 0$, $\mu_s > 0$; and the graviton mass induced by the shear sector is 10^{-42} eV-scale, negligible. What the action *newly defines rather than solves*: the shear fraction w and the medium's Cassini quadrupole Q_2 are now boundary-value computations of a written theory — no longer posits — with the computations specified. The MOND sign is the sign of the medium's compressive response and remains an input; a_0 's value and Z remain inputs; this is Branch B *written*, not adopted, and not a theory of everything.

1. The construction

Kinematics (standard relativistic elasticity). Three scalars $\varphi^a(x)$ label the medium's elements; the internal metric is $B^{ab} = g^{\mu\nu}\partial_\mu\varphi^a\partial_\nu\varphi^b$. The number density is $n = n_0\sqrt{\det B^{ab}}$ (with n_0 the relaxed density), the medium four-velocity u^μ is the unique timelike direction with $u^\mu\partial_\mu\varphi^a = 0$, the **volumetric strain** is $J = -\frac{1}{2}(\det B/\det \bar{B})$ relative to the relaxed configuration $\bar{B}$, and the **shear invariant** is $\Sigma = [\frac{1}{2}\tilde{B} - 3]$ built from the unimodular part $\tilde{B}^{ab} = B^{ab}/(\det B)^{1/3}$ (for small strains, $\Sigma \rightarrow \frac{1}{2}e_{ij}e^{ij}$, the squared deviatoric strain — the J_2 of the campaign lemmas). Internal $SO(3)$ symmetry among the φ^a enforces isotropy; the two invariants (J, Σ) are the *only* independent scalars to this order — the two-invariant structure that decouples the monopole (bulk, $\ell = 0$) from the quadrupole (shear, $\ell = 2$), which the campaign proved scalar AQUAL cannot do.

Dynamics. $S_{\text{med}} = -\int \sqrt{-g}\rho_\Lambda c^2 \mathcal{E}(J, \Sigma)$ with $\mathcal{E} = 1 + [V(J) + \mu_s\Sigma]/\rho_\Lambda c^2$ and:

- **The relaxed point is the dark energy.** $V(0) = 0$, $\Sigma = 0$: the medium contributes $T_{\mu\nu} = -\rho_\Lambda c^2 g_{\mu\nu}$ — an equation-of-state $w_{\text{eos}} = -1$ fluid. The medium does not *add* a dark sector; it *is* the dark energy, deformed.
- **The volumetric sector carries the MOND response.** For $0 < J < J_c$ the potential has the scale-free branch $V'(J) = 2K_{\text{eff}}\sqrt{J/\kappa^2}$ -normalized such that the static response to a baryonic source reproduces the displacement law $g_D = \sqrt{a_0 V g_{\text{bar}}/6}$, with the total deep stiffness (volumetric plus the shear share fixed by the compatibility lemma $J_2 = J_1^2/6$) equal to $K_{\text{eff}} = a_0^2/16\pi G = 2.6 \times 10^{-12}$ Pa. This is the $\sqrt{Z/6} = 0.982$ match, now a property of V .
- **The budget boundary.** The domain of V ends at J_c (equivalently the response terminates at $y_c = Z/2$): implemented as the constrained sector $J \leq J_c$ with multiplier — *not* as a feature of V itself, honoring the impossibility theorem that no single-valued constitutive law can carry the tail. Beyond saturation the medium's displaced entropy budget is exhausted and its response depletes as derived ($T = \min(1, y_c/y)$, the reading supported by Verlinde's text, with the cap alternative computed dead).
- **The shear sector is near-linear by construction and bounded by theorem.** $\mu_s\Sigma$ with constant μ_s : transverse phonons at $v_T = \sqrt{\mu_s/\rho_\Lambda}$. The exact refusal $\rho_\Lambda c^2/K_{\text{eff}} = 6Z^2$ caps $v_T \leq 0.17c$ if the deep amplitude is to survive — the medium's shear waves are slow material sound. The *anharmonic* shear fraction w (the residual coupling of shear to the volumetric nonlinearity through det-geometry) is no longer a free parameter of a prescription: it is a **computable functional of this action**, $w[V, \mu_s]$, via the second-order expansion around the galactic pre-strain state — the campaign's band $w \sim [0.08, 2.7]$ is the pre-action estimate; the action defines the exact object.

Couplings. Baryons couple to $g_{\mu\nu}$ only; photons couple to $g_{\mu\nu}$ only; the medium couples to $g_{\mu\nu}$ only. One metric. Light bends because the deformed medium's $T_{\mu\nu}$ *gravitates* — a genuine source, no

disformal structure, no second cone: GW170817 is satisfied identically ($c_\gamma = c_{\text{GW}} = c$), which is the constraint that created Branch B in the first place.

2. Machine-verified consequences (committed script)

1. **Static weak field (corrected in v2).** The medium's own stress-energy is $10^{6.5-7}$ too small to source the halo (audited: $K_{\text{eff}}\varepsilon^2/c^2 \lesssim 3 \times 10^{-29} \text{ kg m}^{-3}$ vs the required 3×10^{-22} at 10 kpc) – v1 asserted this item without checking it, reproducing the program's earlier bath-energetics kill at the medium's own strain energy. The corrected sourcing is the **nonminimal sector**: linearizing $[1 + h(J)]R$ gives $\nabla^2 \Phi = 4\pi G \rho_b + (c^2/2)\nabla^2 h$, so $\rho_D = (c^2/8\pi G)\nabla^2 h$; matching fixes $h = (2/c^2) \times$ the MOND excess potential, i.e. $h \sim 10^{-6}$ (a weak dressing of GR), with the deep-branch closed form $h \propto (\sqrt{a_0 GM}/c^2) \ln r$ and $h(J)$ obtained by eliminating r on the matched background. High- g : $\nabla h \propto (\nu - 1)g \rightarrow a_0/2$ – the same deep-Newton suppression the modified-inertia evasion used, so the h -channel monopole is solar-system-safe.
2. **Cosmology and $a_0(z)$.** On FRW the relaxed medium gives $\rho_{\text{DE}} = \rho_\Lambda$, $w_{\text{eos}} = -1$; promoting the relaxed density to the measured dark-energy density, the deep stiffness scales as $K_{\text{eff}}(z) = a_0(z)^2/16\pi G$ with $a_0(z) = \frac{c}{2}\sqrt{G\rho_{\text{DE}}(z)}$ – the program's one distinctive falsifiable prediction now follows from the action's own normalization instead of being separately postulated. If dark energy is a true cosmological constant, a_0 is constant and the distinctive content dissolves, exactly as pre-registered.
3. **Degrees of freedom and stability.** Three phonons (one longitudinal, $c_L^2 = (V'' + \frac{4}{3}\mu_s)/\rho_\Lambda$; two transverse, $c_T^2 = \mu_s/\rho_\Lambda$), ghost-free and gradient-stable for $V'' > 0$, $\mu_s > 0$; the metric tensor sector is standard GR (two polarizations at c); the shear-induced graviton mass $m_{\text{GW}}^2 \sim 16\pi G\mu_s/c^4$ is of order the Hubble scale squared – utterly negligible against all bounds.
4. **The exact numbers carried.** $K_{\text{eff}} = a_0^2/16\pi G$, $\rho_\Lambda c^2/K_{\text{eff}} = 6Z^2 = 201.06$, $v_T^{\text{max}} = c\sqrt{6}/(6Z^2)^{1/2}$ -normalized $\leq 0.17c$, $y_c = Z/2 = 2.894$ – all reproduced symbolically from the action's own definitions, both footings.

3. What the action newly defines (the work it creates)

The point of writing the theory is that Branch B's two remaining physics unknowns stop being posits and become **computations of a specific Lagrangian**: **(a)** the anharmonic shear fraction – expand $\mathcal{E}$ to second order in the probe strain around the galactic pre-strain solution of *this* action and read off w (the campaign's straddling band $[0.08, 2.7]$ becomes one number, and the strain normalization κ that made w convention-hostage is fixed by the action's own J -definition); **(b)** the medium's true Cassini Q_2 – the axisymmetric boundary-value problem for φ^a around the Sun in the galactic external field, with the $\ell = 0/\ell = 2$ decoupling now exact rather than argued. Both are finite classical computations. Their outcomes are genuinely open: if w computes small, Branch B clears Cassini and stands as a complete, GW-safe, lensing-correct field theory of the dark sector at the price of the inputs (s , a_0 , Z); if w computes large, Branch B is dead by its own action and the program reverts to Branch A with lensing an open deficit. The action makes the branch *falsifiable by calculation*.

4. Honest ledger

New here: the covariant action itself – relativistic elastic *linear* dark energy is prior art [Bucher & Spergel 1999; Battye & Moss 2007], and the solid-EFT kinematics are standard [Endlich–Nicolis–Wang 2013]; what has not previously been written is the **Verlinde-class anharmonic volumetric sector** (the scale-free $\sqrt{J}$ branch matched to $a_0 = c^2\sqrt{\Lambda/32\pi}$) **with the entropy-budget domain**

boundary at the derived $y_c = Z/2$ — the constitutive content that makes an elastic solid produce MOND-scale phenomenology. **Inherited, not new:** every matching coefficient (0.982, $y_c = Z/2$, $6Z^2$, K_{eff}) comes from the previously published, machine-verified campaign results — nothing is re-fitted here. **Still inputs:** the MOND sign, a_0 's value, Z ; and (new, honest) the *form* of $h(J)$ is matched to the displacement law, not derived — the constitutive debt moves from the (refuted) minimal sourcing claim into the nonminimal coupling, and the theory is now explicitly scalar-tensor-like: Branch B is modified gravity with a medium, as the program's ledger has always stated. **Still open:** the two defined computations of §3; the medium's full nonlinear well-posedness (the constrained sector's dynamics at the budget boundary); cluster-scale performance; and the Branch A/B identity choice — *writing* Branch B's action is not *adopting* it, and the observational fingerprint distinguishing it from Branch A remains indistinguishable at current sensitivity. No completeness or theory-of-everything claim is made.

Reproducibility

real_research/reviews/shear_linearity_2026/elastic_action_consequences.py (items 2–6), action_source_audit_and_fix.py (the v2 correction: the minimal-sourcing audit and the nonminimal weak-field derivation), action_w_q2_computation.py (the pinned κ , reconstructed (S, f_1) , and the w band $[0.13, 0.53]$ — one lever, the anisotropic-BVP kernel, remaining) plus the campaign scripts previously committed (p6_pin_verlinde_coefficients.py, p6_throttle_fingerprint_sparc.py, lane1_theorem.py, ../branchB_q2_gate_2026/, ../lensing_source_fork_2026/). Program concept DOI: 10.5281/zenodo.21253644; the $y_c = Z/2$ note: 10.5281/zenodo.21300855.

References

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Both a_0 footings; every coefficient inherited from committed, verified computations; Branch B written, not adopted.